# MATH 160 — Counting Methods

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Name: _____ Date: _____ Score: / 10

**Learning Objectives**

- Apply the Multiplication Rule for sequential choices
- Compute permutations P(n,r) when order matters
- Compute combinations C(n,r) when order does not matter
- Use the Pigeonhole Principle to solve counting arguments

*Always ask: does ORDER matter? If yes → permutation. If no → combination. For multi-step problems, multiply counts for each independent step.*

**1. A restaurant offers 4 appetizers, 6 entrees, and 3 desserts. How many different 3-course meals are possible? (Multiplication Rule)**

$$4 \times 6 \times 3 = ?$$

Answer: _____

**2. In how many ways can 6 students line up for a photo? (Permutation — all n objects)**

$$6! = ?$$

Answer: _____

**3. A club must elect a President and Vice-President from 10 members. How many ordered outcomes are possible? (Permutation P(n, r))**

$$P(10, 2) = \frac{10!}{(10-2)!}$$

Answer: _____

**4. A committee of 4 is chosen from 9 people. How many committees are possible? (Combination C(n, r))**

$$C(9, 4) = \frac{9!}{4!\ 5!}$$

Answer: _____

**5. A poker hand of 5 cards is dealt from a standard 52-card deck. How many hands contain exactly 3 Aces?**

$$C(4, 3) \times C(48, 2) = ?$$

Answer: _____



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**6. How many 5-character passwords can be formed using lowercase letters (a–z) if repetition is allowed? If repetition is NOT allowed?**

$26^5$ vs $P(26, 5)$

Answer: _____

**7. By the Pigeonhole Principle: If 13 socks are drawn from a drawer containing only red and blue socks, what is the minimum number guaranteed to have at least 7 socks of the same color?**

Pigeonhole: $n$ items, $k$ holes $\Rightarrow \lceil n/k \rceil$

Answer: _____

**8. How many ways can 8 books be arranged on a shelf if 3 specific books must stay together?**

$6! \times 3! = ?$

Answer: _____

**9. How many different 4-digit numbers can be formed using digits 1–9 with no digit repeated, and the first digit must be odd?**

$5 \times P(8, 3) = ?$

Answer: _____

**10. Expand C(n, r) = C(n, n–r) to show that C(10, 7) = C(10, 3). Compute both.**

$$C(10, 7) = C(10, 3) = \frac{10!}{3!\ 7!}$$

Answer: _____



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*Problems 1–2: multiplication rule and factorial. Problems 3–4: P(n,r) vs C(n,r) contrast. Problem 7 (Pigeonhole): set up "worst case" scenario first.*

## Solutions

**1.** A restaurant offers 4 appetizers, 6 entrees, and 3 desserts. How many different 3-course meals are possible? (Multiplication Rule)

$4 \times 6 \times 3 = ?$

→ *Multiplication Rule: choices for each independent step multiply.*
→ *4 × 6 × 3 = 72 different 3-course meals.*

**Answer:** $4 \times 6 \times 3 = 72$ meals

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**2.** In how many ways can 6 students line up for a photo? (Permutation — all n objects)

$6! = ?$

→ *All 6 students are arranged in a row — order matters.*
→ *6! = 6 × 5 × 4 × 3 × 2 × 1 = 720.*

**Answer:** $6! = 720$

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**3.** A club must elect a President and Vice-President from 10 members. How many ordered outcomes are possible? (Permutation P(n, r))

$$P(10, 2) = \frac{10!}{(10-2)!}$$

→ *P(n,r) = n!/(n–r)! — order matters (President ≠ Vice-President).*
→ *P(10,2) = 10!/8! = 10 × 9 = 90.*

**Answer:** $P(10, 2) = 10 \times 9 = 90$

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**4.** A committee of 4 is chosen from 9 people. How many committees are possible? (Combination C(n, r))

$$C(9, 4) = \frac{9!}{4!\ 5!}$$

→ *C(n,r) = n!/(r!(n–r)!) — order does NOT matter.*
→ *C(9,4) = 9!/(4!·5!) = (9×8×7×6)/(4×3×2×1) = 3024/24 = 126.*

**Answer:** $C(9, 4) = 126$

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**5.** A poker hand of 5 cards is dealt from a standard 52-card deck. How many hands contain exactly 3 Aces?

$C(4, 3) \times C(48, 2) = ?$

→ *Choose 3 Aces from 4: C(4,3) = 4.*
→ *Choose the remaining 2 cards from the 48 non-Aces: C(48,2) = 1128.*
→ *Total = 4 × 1128 = 4512 hands.*

**Answer:** $C(4, 3) \cdot C(48, 2) = 4 \times 1128 = 4512$

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**6.** How many 5-character passwords can be formed using lowercase letters (a–z) if repetition is allowed? If repetition is NOT allowed?

$26^5$ vs $P(26, 5)$

→ *With repetition: 26 choices for each of 5 slots = $26^5$ = 11,881,376.*
→ *Without repetition: P(26,5) = 26×25×24×23×22 = 7,893,600.*

**Answer:** $26^5 = 11{,}881{,}376$ vs $P(26, 5) = 7{,}893{,}600$

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**7.** By the Pigeonhole Principle: If 13 socks are drawn from a drawer containing only red and blue socks, what is the minimum number guaranteed to have at least 7 socks of the same color?

Pigeonhole: $n$ items, $k$ holes ⇒ $\lceil n/k \rceil$

→ *Worst case: 6 red and 6 blue = 12 socks with no color having 7.*
→ *The 13th sock must be one of the two colors → one color has ≥ 7.*
→ *Pigeonhole: $\lceil 13/2 \rceil = 7$. Guaranteed after drawing 13.*

**Answer:** 13 socks, 2 colors ⇒ ≥ 7 of one color

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**8.** How many ways can 8 books be arranged on a shelf if 3 specific books must stay together?

$6! \times 3! = ?$

→ *Treat the 3 books as ONE block → 6 objects total (block + 5 other books).*
→ *Arrange 6 objects: 6! = 720 ways.*
→ *Arrange the 3 books WITHIN the block: 3! = 6 ways.*
→ *Total = 720 × 6 = 4320.*

**Answer:** $6! \times 3! = 720 \times 6 = 4320$

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**9.** How many different 4-digit numbers can be formed using digits 1–9 with no digit repeated, and the first digit must be odd?

$5 \times P(8, 3) = ?$

→ *First digit: must be odd (1,3,5,7,9) → 5 choices.*
→ *Remaining 3 digits from 8 remaining (no repetition): P(8,3) = 8×7×6 = 336.*
→ *Total = 5 × 336 = 1680.*

**Answer:** $5 \times 8 \times 7 \times 6 = 1680$

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**10.** Expand C(n, r) = C(n, n–r) to show that C(10, 7) = C(10, 3). Compute both.

$$C(10, 7) = C(10, 3) = \frac{10!}{3!\ 7!}$$

→ *C(10,7) = 10!/(7!·3!) = C(10,3) — same calculation.*
→ *C(10,3) = (10×9×8)/(3×2×1) = 720/6 = 120.*
→ *Symmetry: choosing 7 to include = choosing 3 to exclude.*

**Answer:** $C(10, 3) = 120$